In [1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

In [12]:
```
data = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/m
```

In [13]:
```
data.shape
```

Out[13]:
```
(891, 12)
```

In [16]:
```
data.describe()
```

Out[16]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329

In [17]:
```
data.describe(include='object')
```

Out[17]:

	Name	Sex	Ticket	Cabin	Embarked
count	891	891	891	204	889
unique	891	2	681	147	3
top	Dooley, Mr. Patrick	male	347082	G6	S
freq	1	577	7	4	644

In [19]:
```
data.isnull().sum()
```

Out[19]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [20]:
```
data['Age'] = data['Age'].fillna(data['Age'].mean())
```

In [23]:
```
data['Cabin'] = data['Cabin'].fillna(data['Cabin'].mode()[0])
```

In [25]:
```
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])
```

In [26]:
```
data.isnull().sum()
```

Out[26]:
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          0
Embarked       0
dtype: int64
```

In [30]:
```
data
```

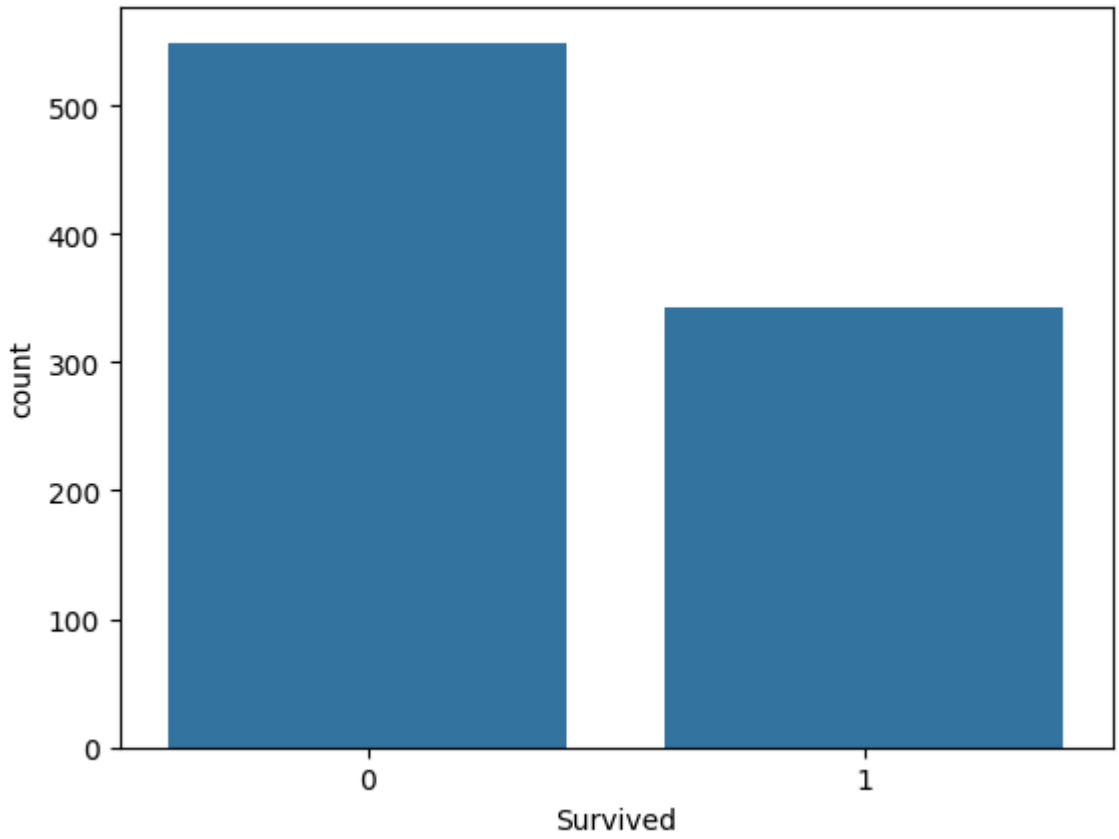
Out[30]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Tick
0	1	0	3	Braund, Mr. Owen Harris	male	22.000000	1	0	A 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.000000	1	0	PC 175
2	3	1	3	Heikkinen, Miss. Laina	female	26.000000	0	0	STON/C 31012
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.000000	1	0	1138
4	5	0	3	Allen, Mr. William Henry	male	35.000000	0	0	3734
...	...	...	...	...	...	...	...	...	
886	887	0	2	Montvila, Rev. Juozas	male	27.000000	0	0	2115
887	888	1	1	Graham, Miss. Margaret Edith	female	19.000000	0	0	1120
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	29.699118	1	2	W. 66
889	890	1	1	Behr, Mr. Karl Howell	male	26.000000	0	0	1113
890	891	0	3	Dooley, Mr. Patrick	male	32.000000	0	0	3703

891 rows × 12 columns

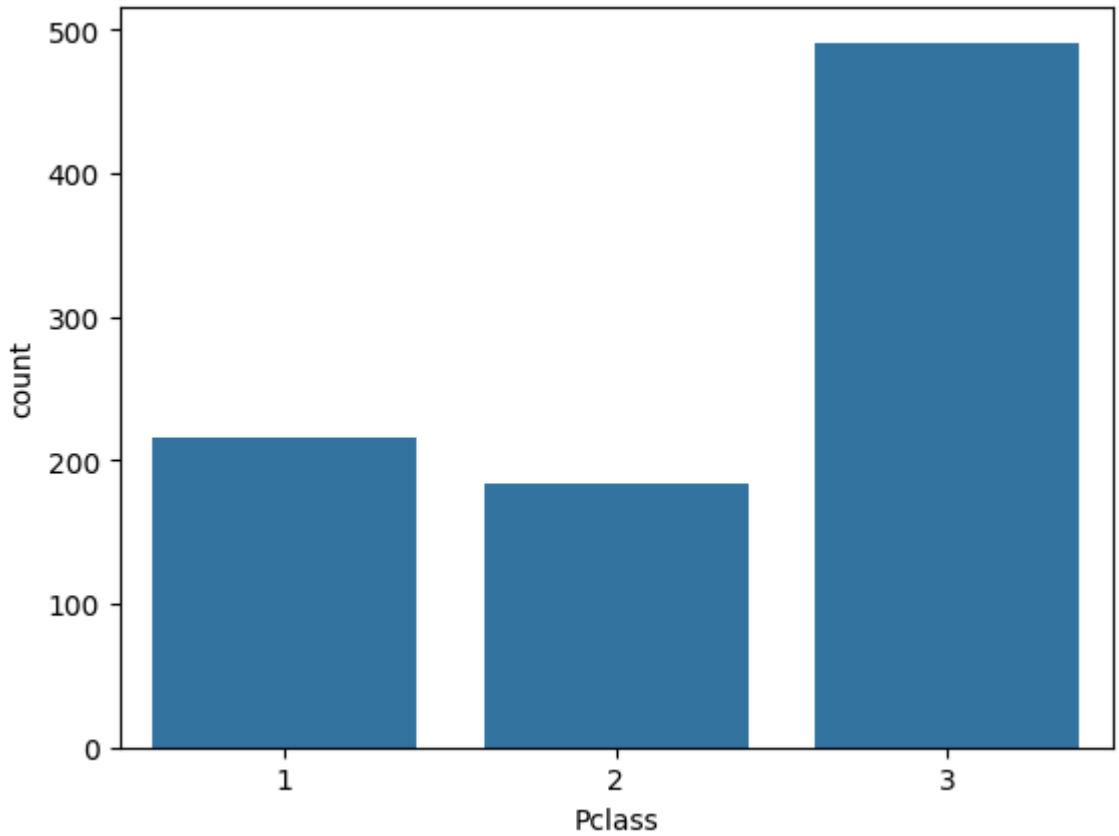
In [41]:
```
sns.countplot(x='Survived', data=data)
```

Out[41]:
```
<Axes: xlabel='Survived', ylabel='count'>
```



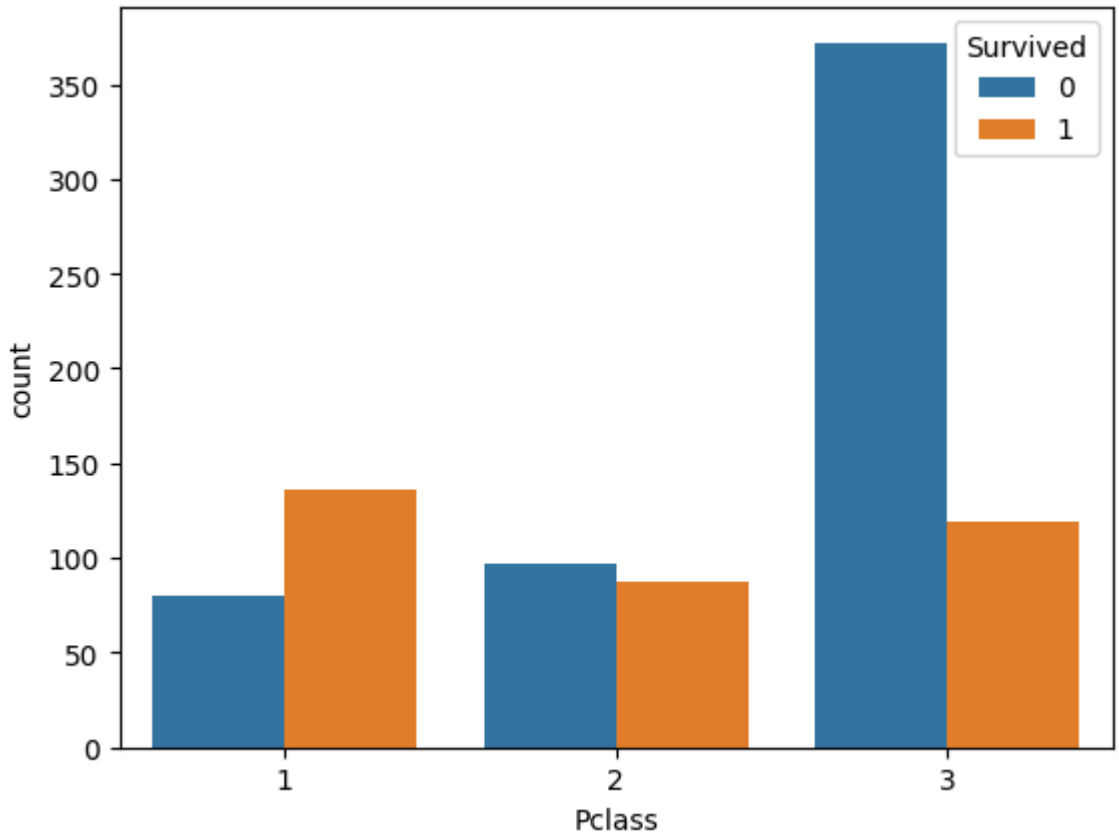
In [42]:
```
sns.countplot(x='Pclass', data = data)
```

Out[42]: <Axes: xlabel='Pclass', ylabel='count'>



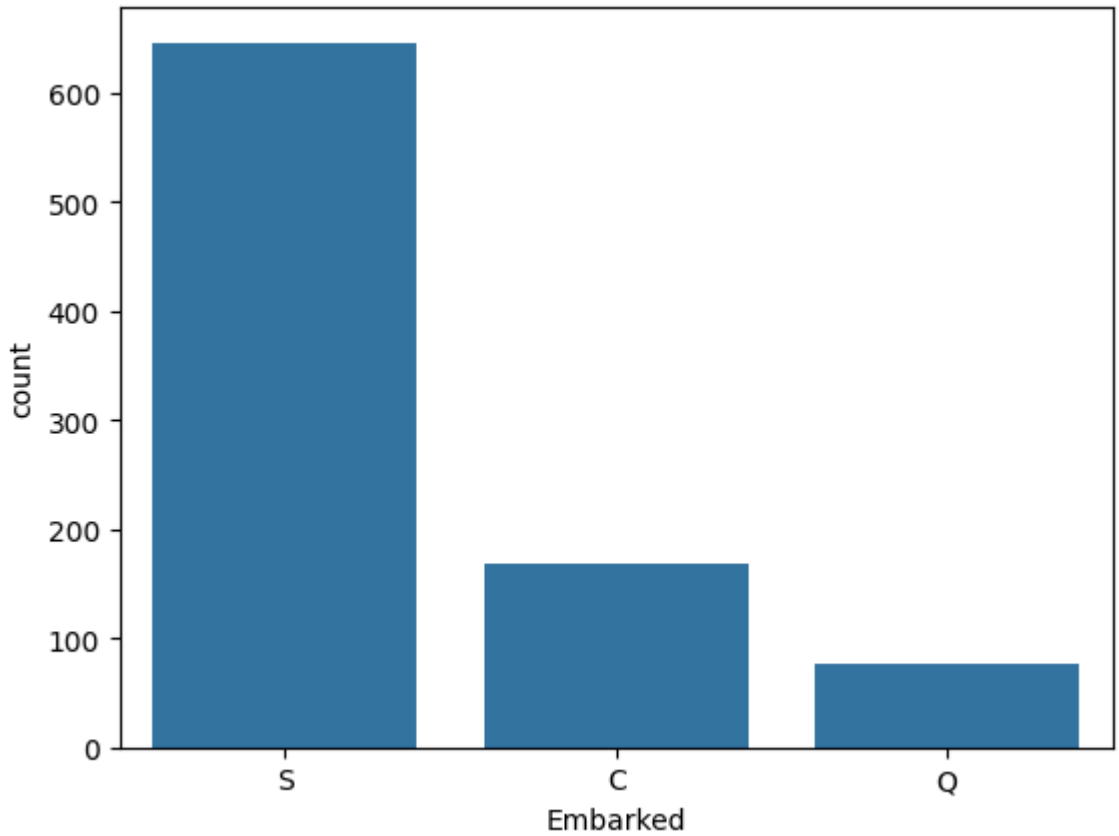
In [43]:
```
sns.countplot(x='Pclass', hue='Survived', data=data)
```

Out[43]: <Axes: xlabel='Pclass', ylabel='count'>



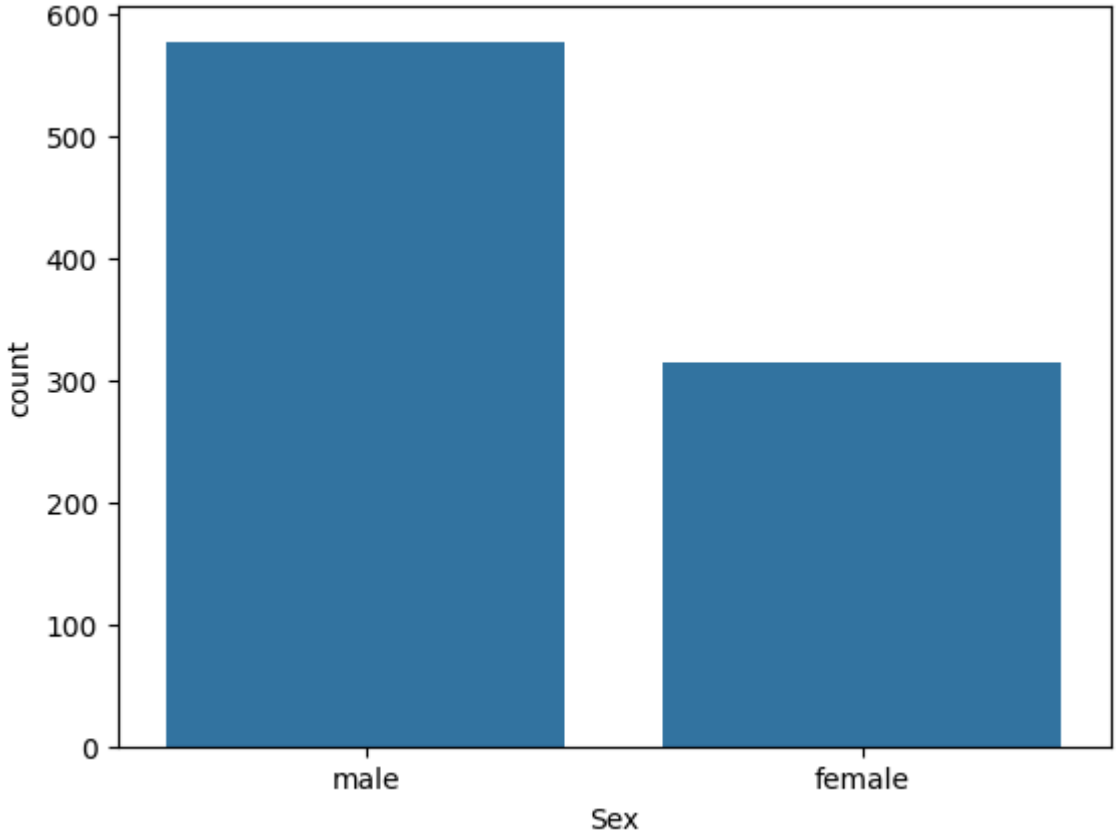
In [44]:
```
sns.countplot(x='Embarked', data=data)
```

Out[44]: <Axes: xlabel='Embarked', ylabel='count'>



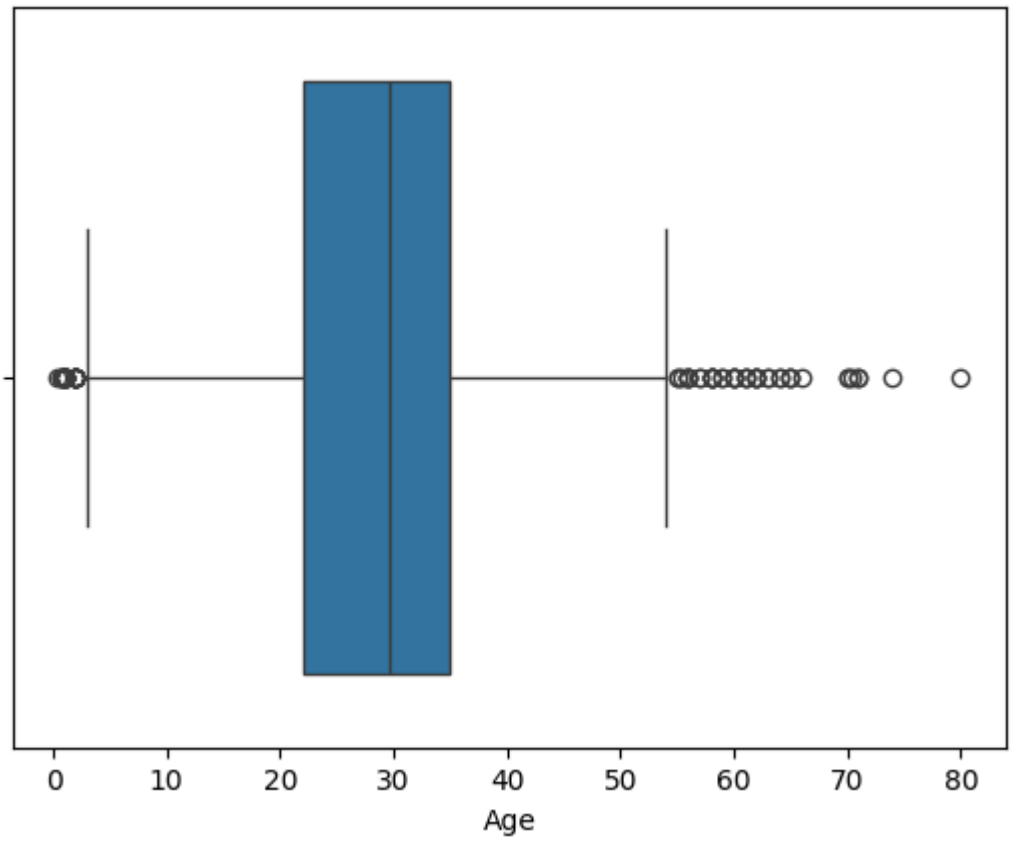
In [47]:
```
sns.countplot(x='Sex', data=data)
```

Out[47]: <Axes: xlabel='Sex', ylabel='count'>



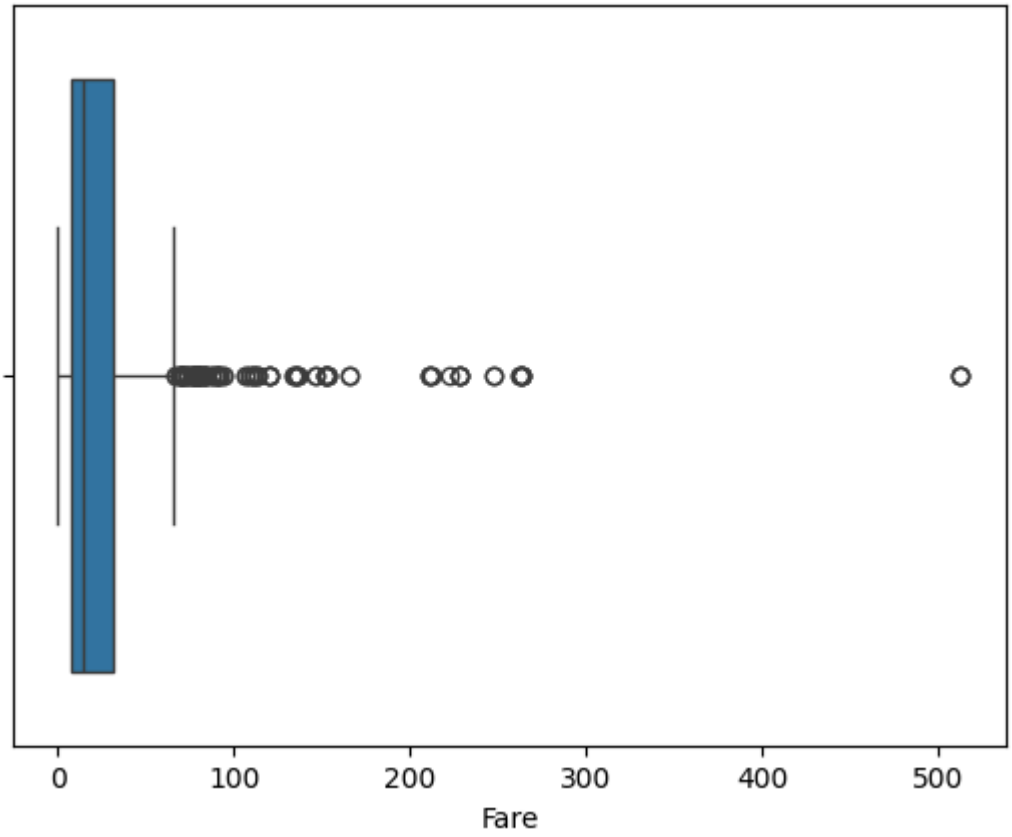
In [54]:
```
sns.boxplot(x='Age', data=data)
```

Out[54]: <Axes: xlabel='Age'>



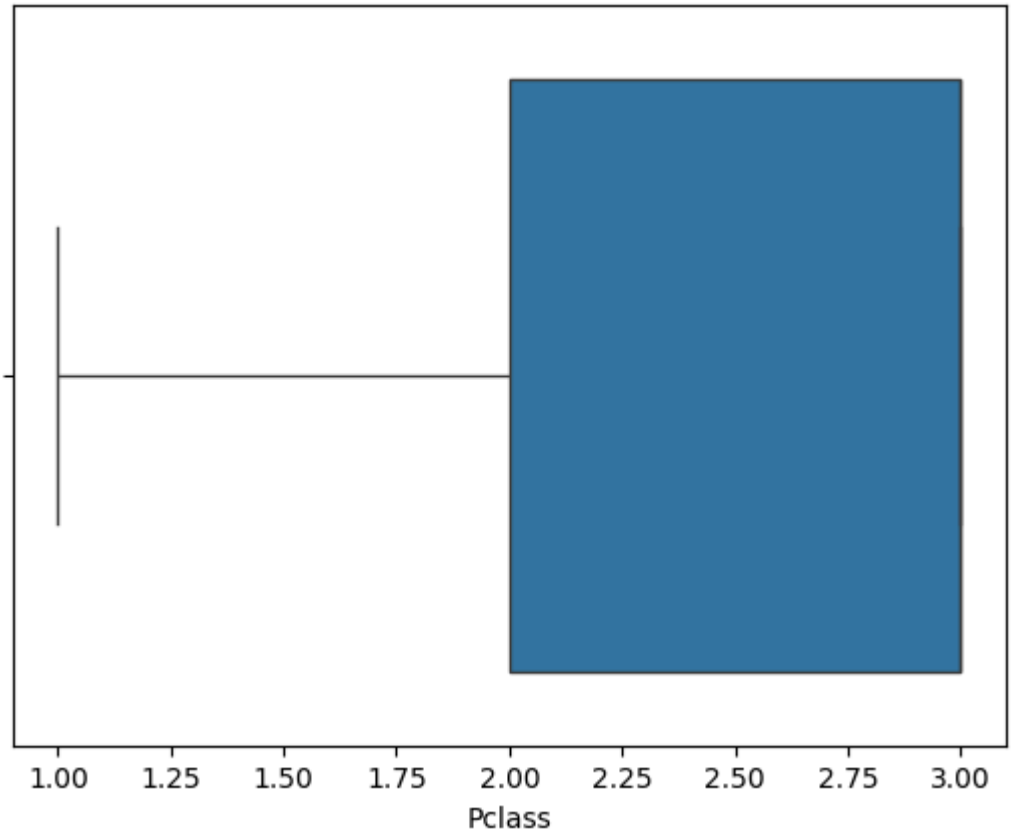
In [55]:
```
sns.boxplot(x='Fare', data=data)
```

Out[55]: <Axes: xlabel='Fare'>



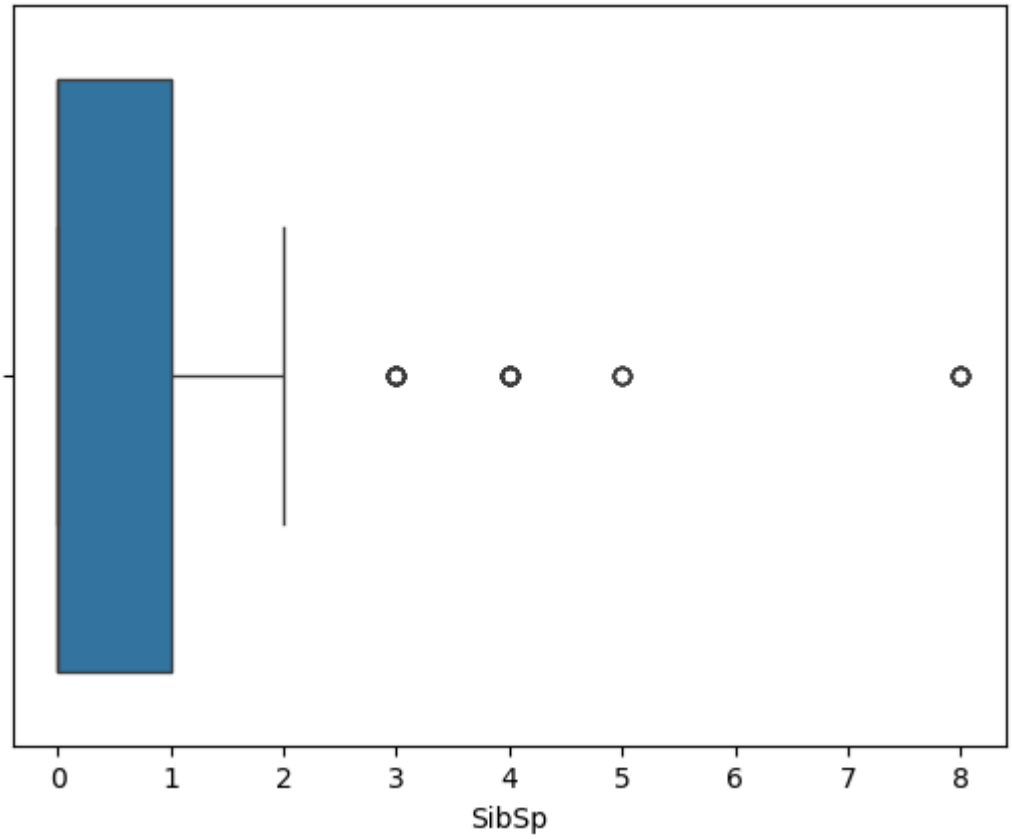
In [56]:
```
sns.boxplot(x='Pclass', data=data)
```

Out[56]:
```
<Axes: xlabel='Pclass'>
```



In [58]:
```
sns.boxplot(x='SibSp', data=data)
```

Out[58]:
```
<Axes: xlabel='SibSp'>
```

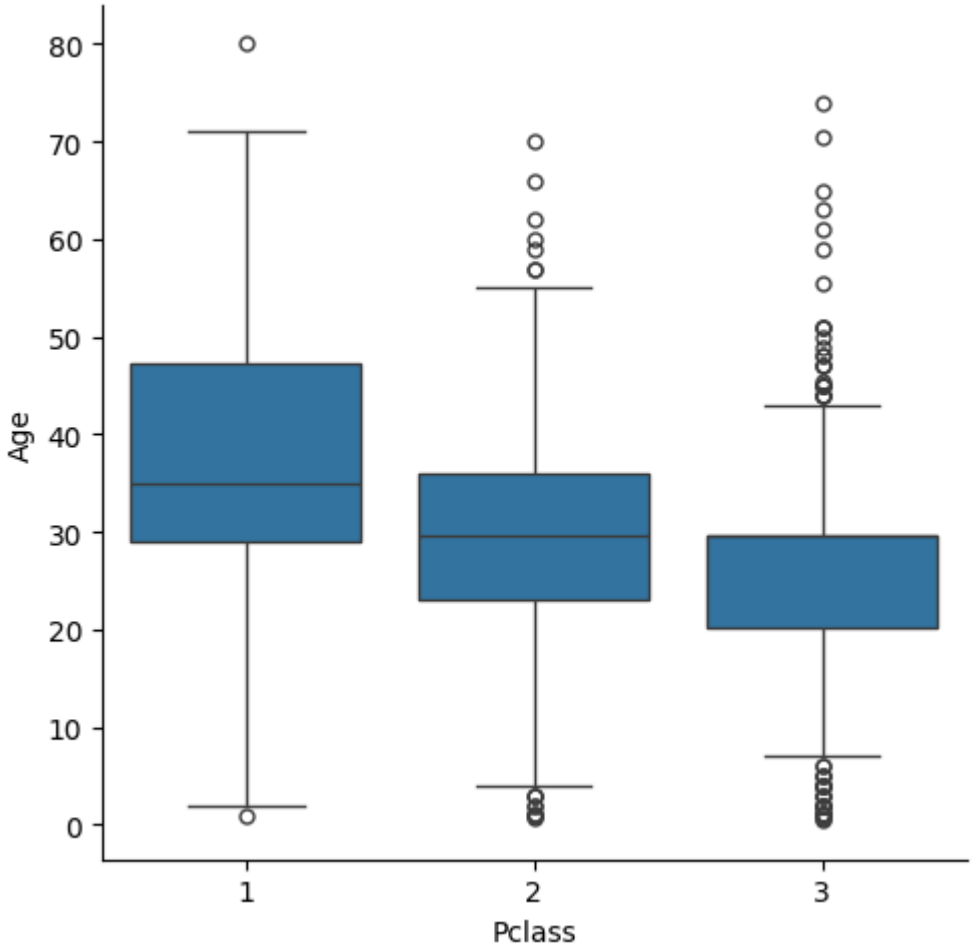


In [59]:
```
data['SibSp'].median()
```

Out[59]:
```
np.float64(0.0)
```

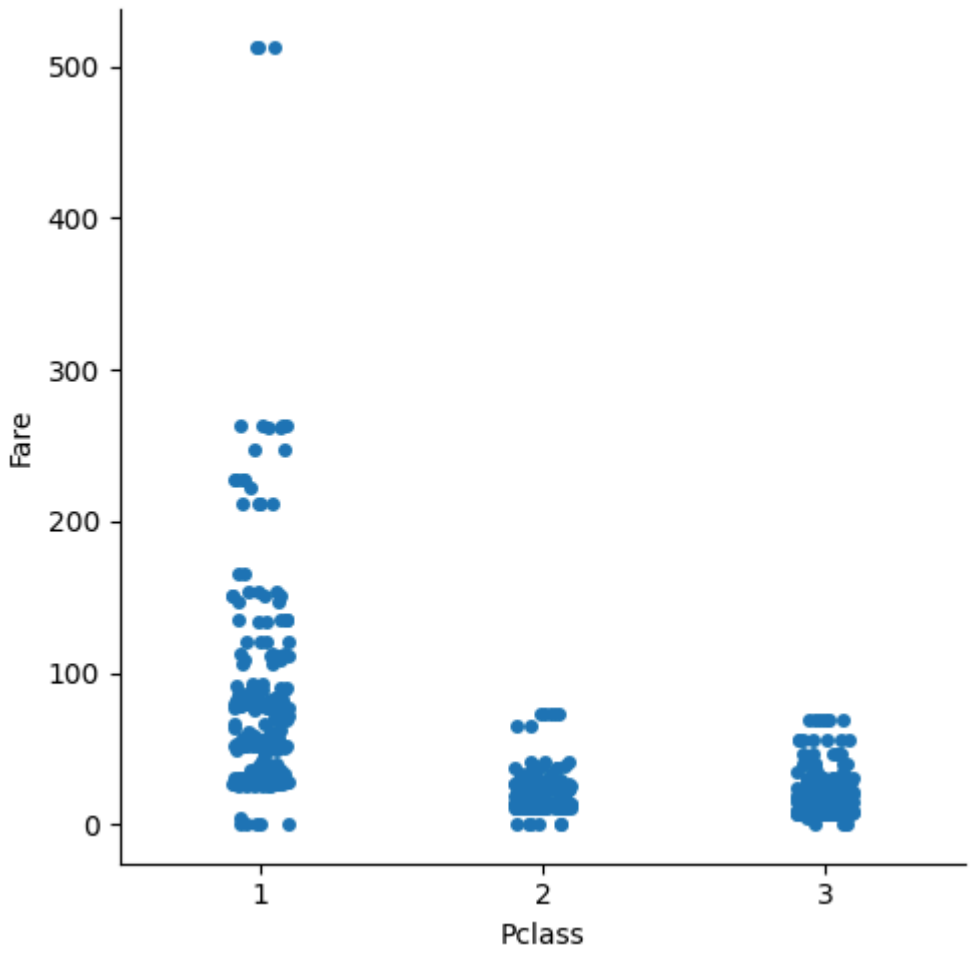
In [60]:
```
sns.catplot(x='Pclass', y='Age', data=data, kind='box' )
```

Out[60]:
```
<seaborn.axisgrid.FacetGrid at 0x1cea9a1bcb0>
```



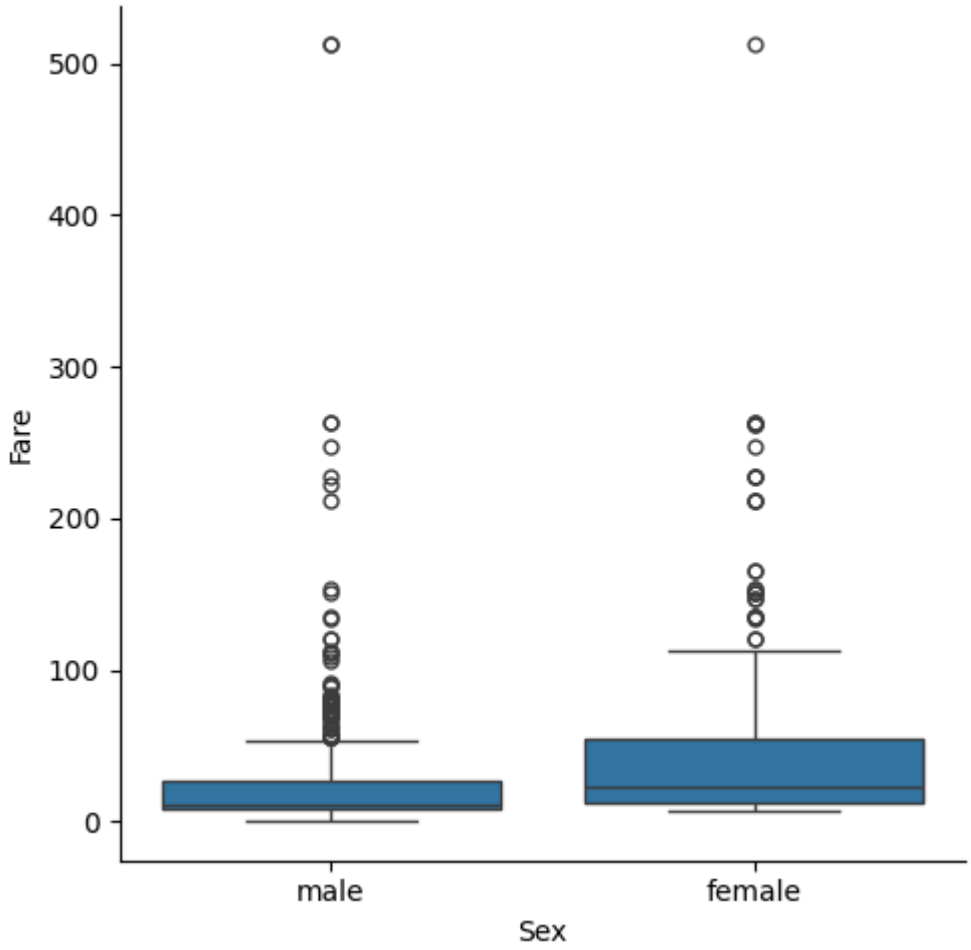
In [61]:
```
sns.catplot(x= 'Pclass', y = 'Fare', data=data, kind = 'strip')
```

Out[61]:
```
<seaborn.axisgrid.FacetGrid at 0x1ceb26e9f90>
```



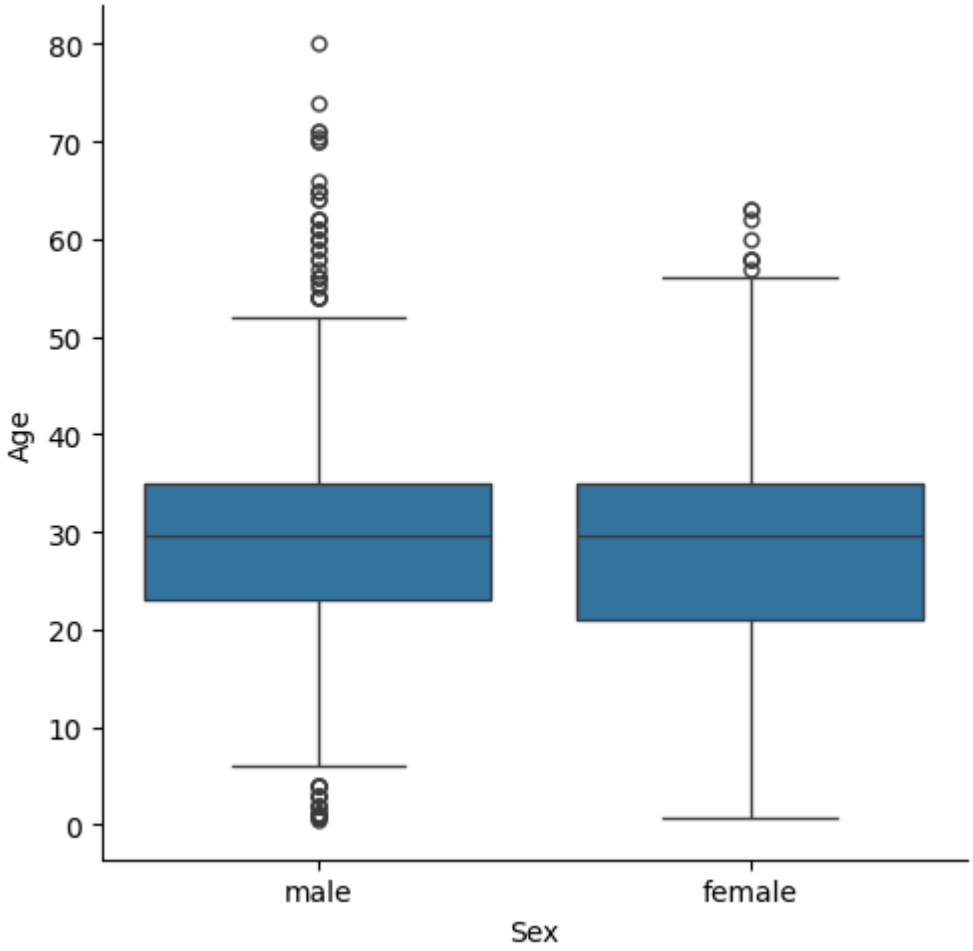
In [64]:
```
sns.catplot(x='Sex', y='Fare', data=data, kind='box')
```

Out[64]:
```
<seaborn.axisgrid.FacetGrid at 0x1ceb2908b90>
```



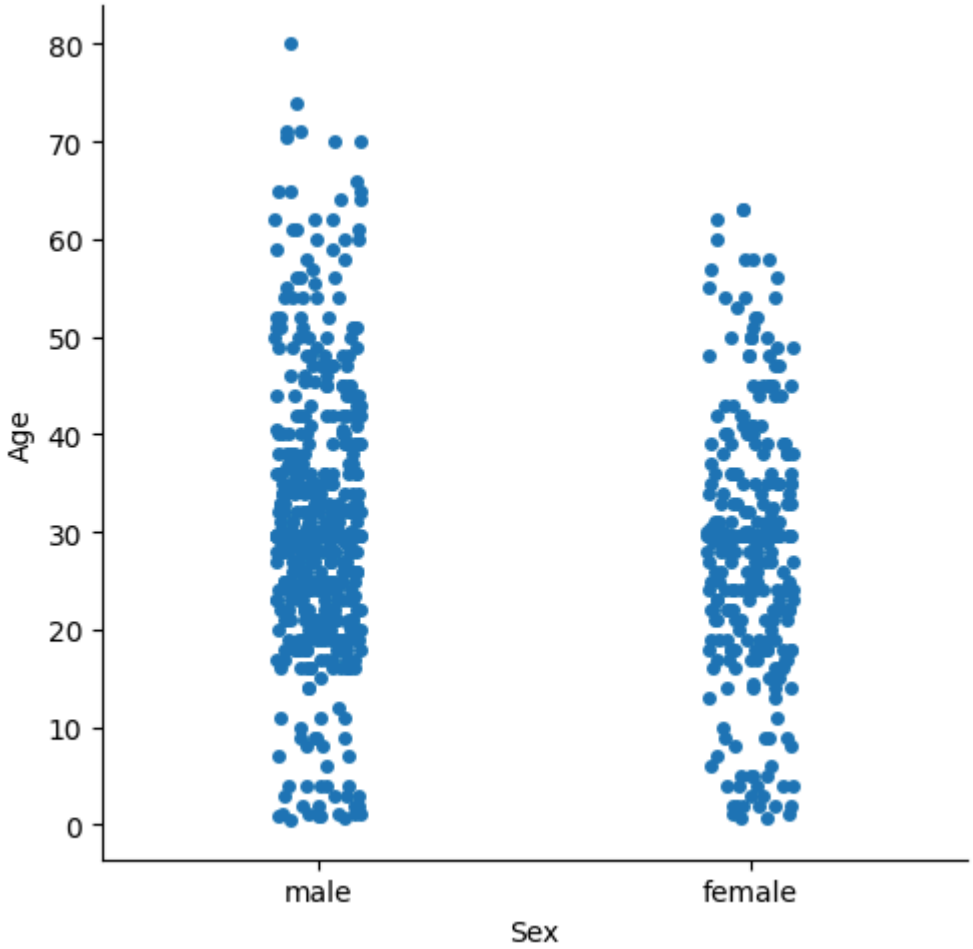
In [65]:
```
sns.catplot(x='Sex', y='Age', data=data, kind='box')
```

Out[65]:
```
<seaborn.axisgrid.FacetGrid at 0x1ceb294f390>
```



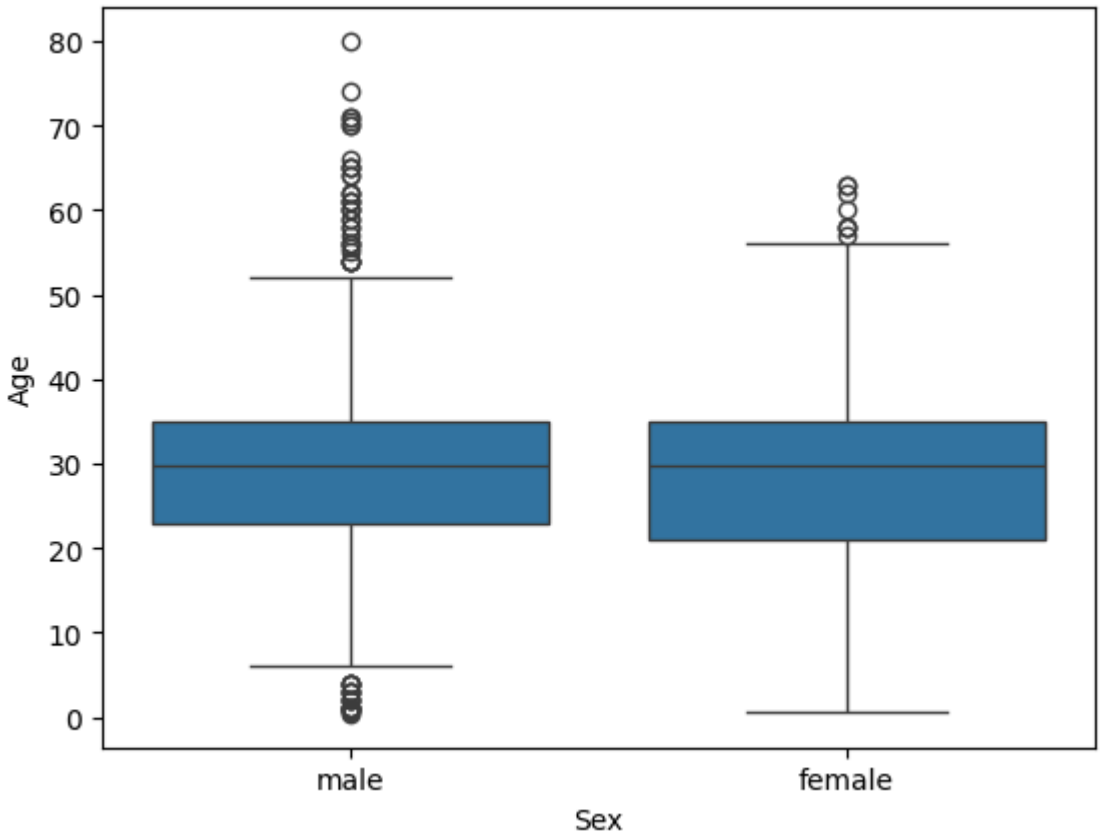
In [67]:
```
sns.catplot(x='Sex', y='Age', data=data, kind='strip')
```

Out[67]:
```
<seaborn.axisgrid.FacetGrid at 0x1ceb3a58b90>
```



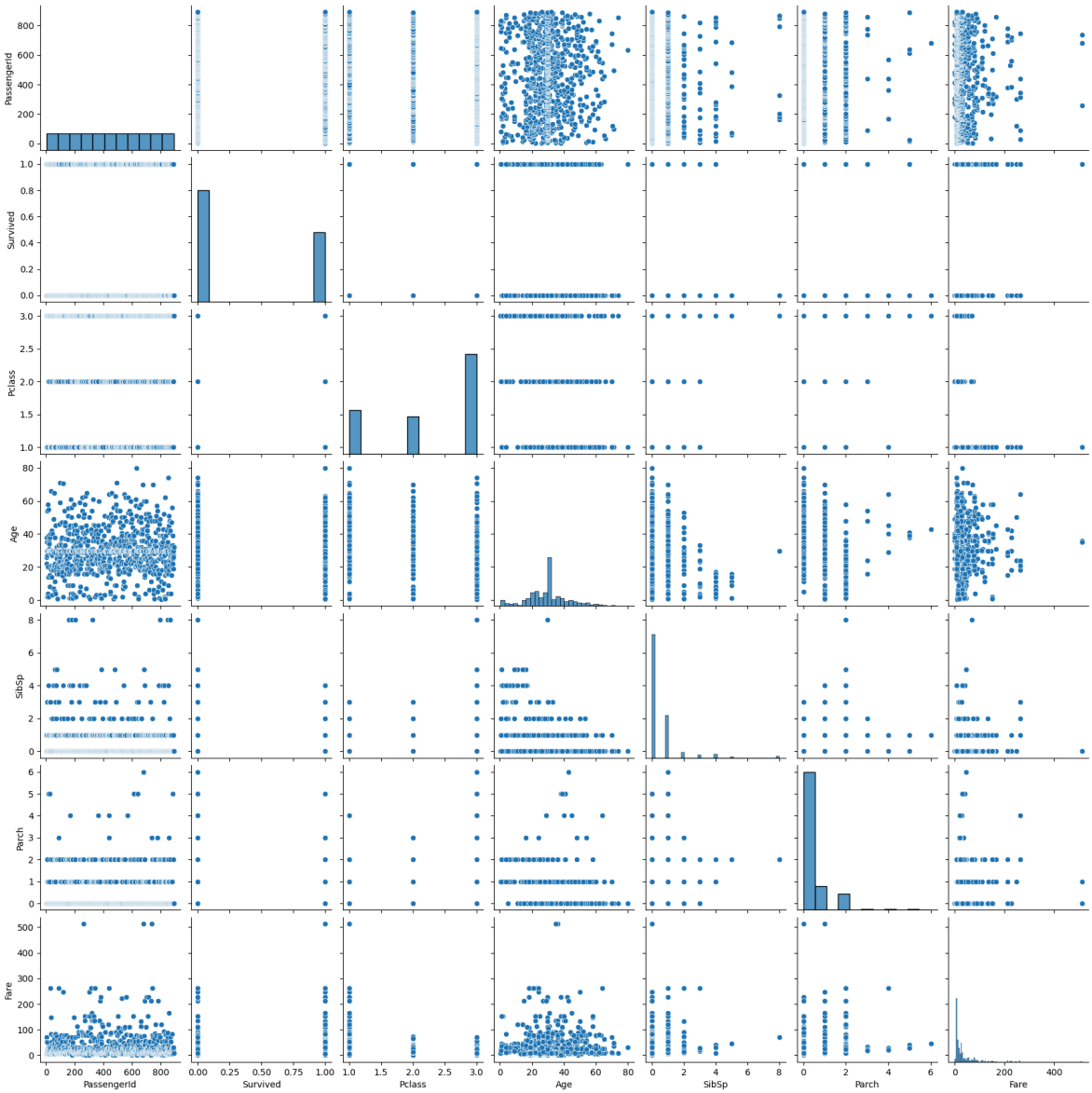
In [69]:
```
sns.boxplot(x='Sex', y='Age', data=data)
```

Out[69]: <Axes: xlabel='Sex', ylabel='Age'>



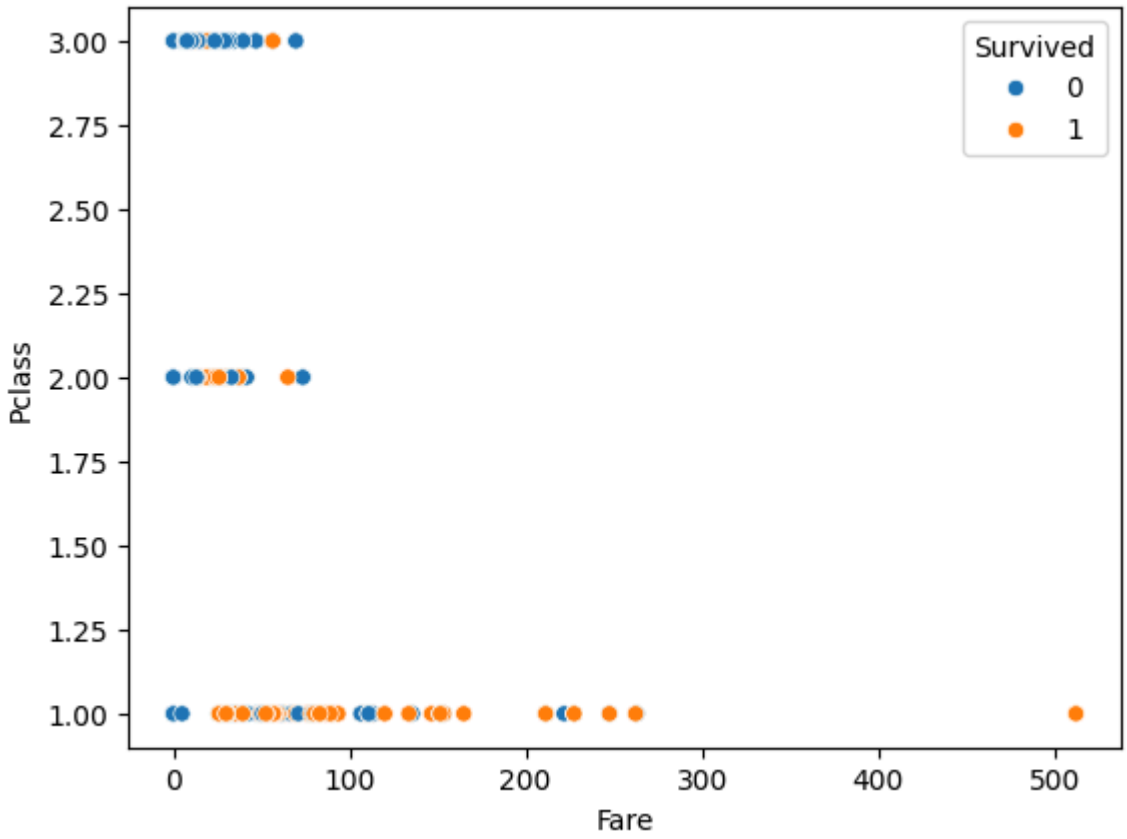
In [70]:
```
sns.pairplot(data)
```

Out[70]:
```
<seaborn.axisgrid.PairGrid at 0x1ceb28a82f0>
```



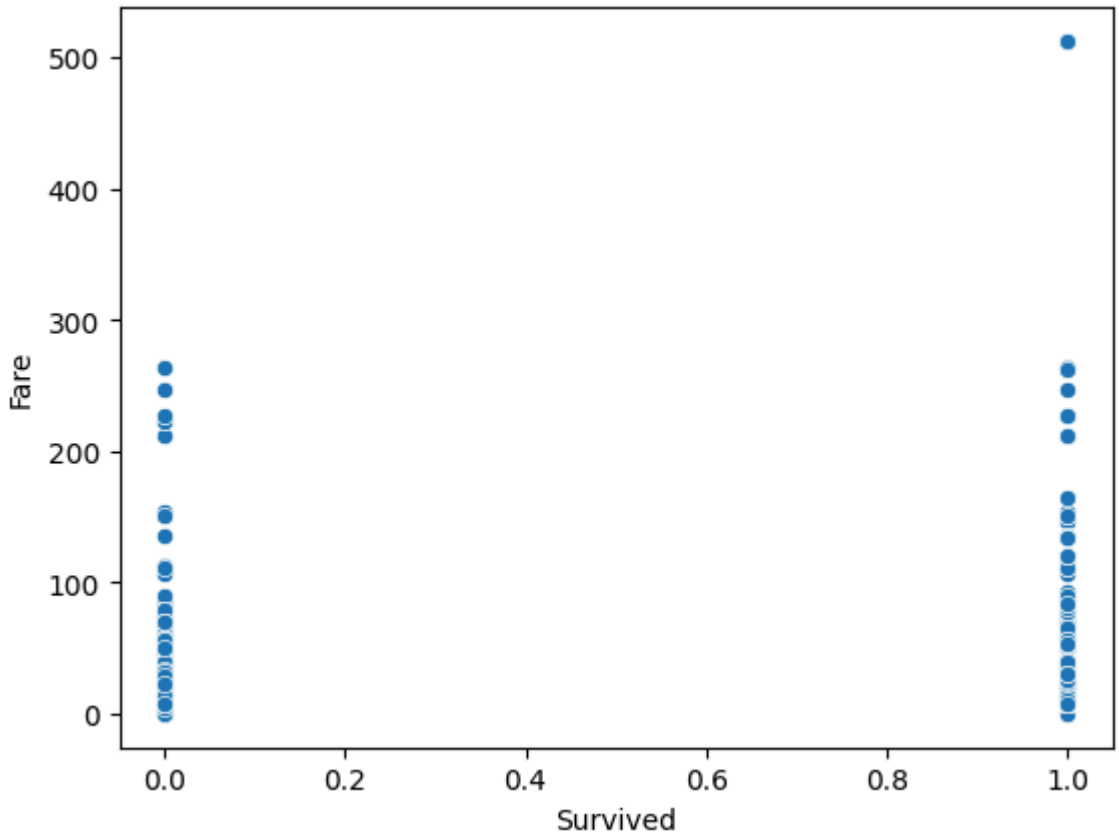
In [73]:
```
sns.scatterplot(x='Fare', y='Pclass', data=data, hue='Survived')
```

Out[73]:
```
<Axes: xlabel='Fare', ylabel='Pclass'>
```



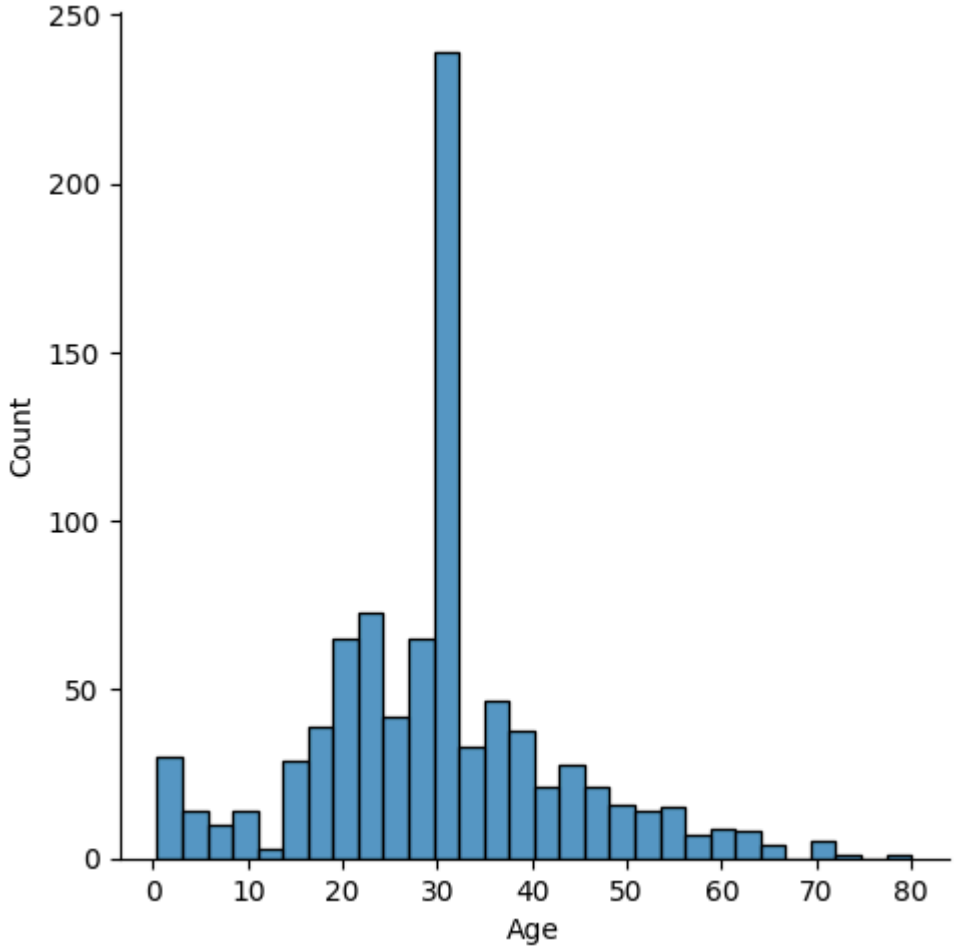
In [74]:
```
sns.scatterplot(x='Survived', y='Fare', data=data)
```

Out[74]:
```
<Axes: xlabel='Survived', ylabel='Fare'>
```



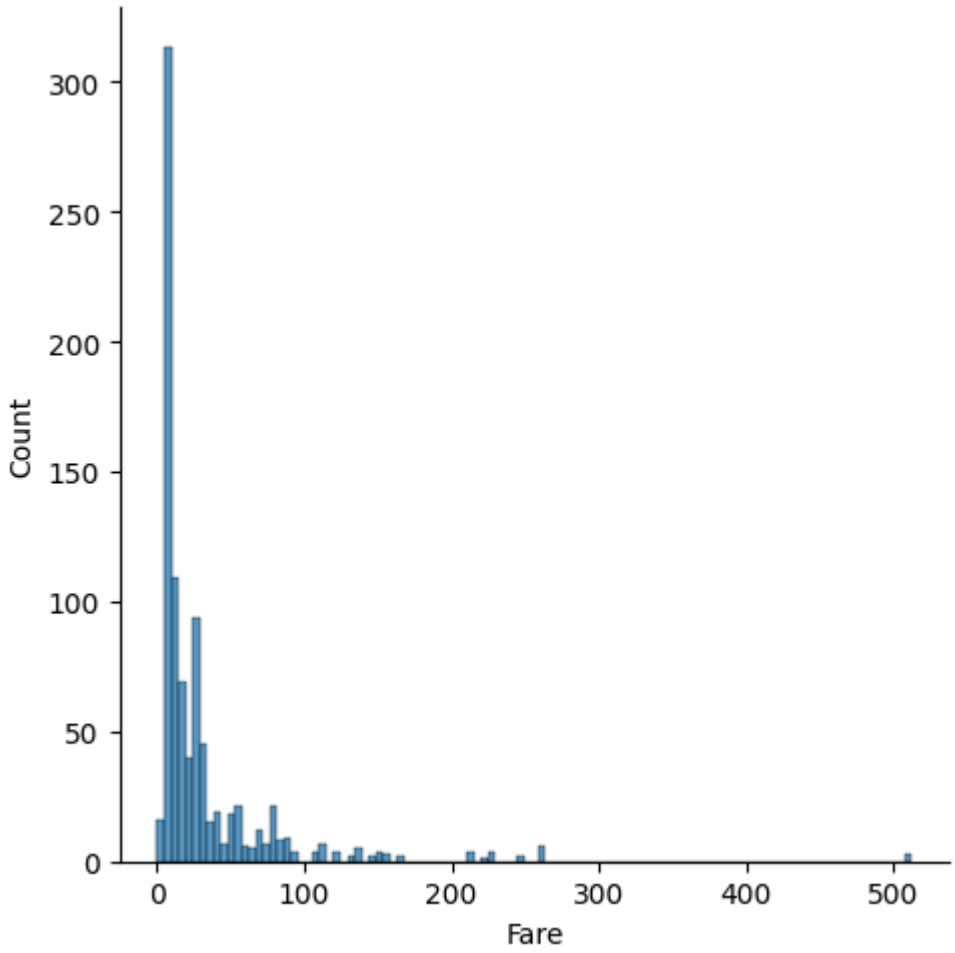
In [76]:
```
sns.displot(data['Age'])
```

Out[76]:
```
<seaborn.axisgrid.FacetGrid at 0x1ceb7a73390>
```



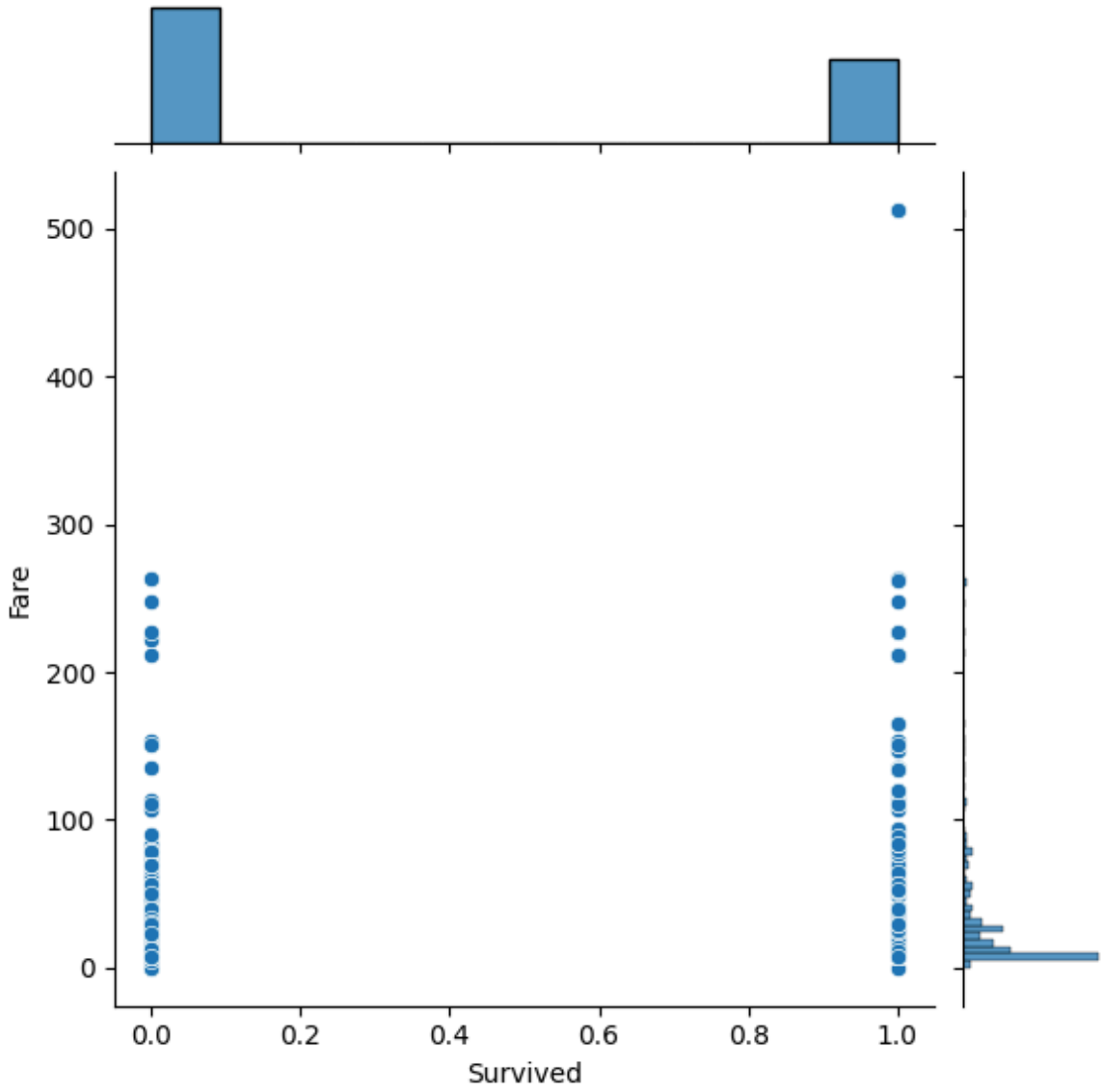
In [78]:
```
sns.displot(data['Fare'])
```

Out[78]:
```
<seaborn.axisgrid.FacetGrid at 0x1ceb9b647d0>
```



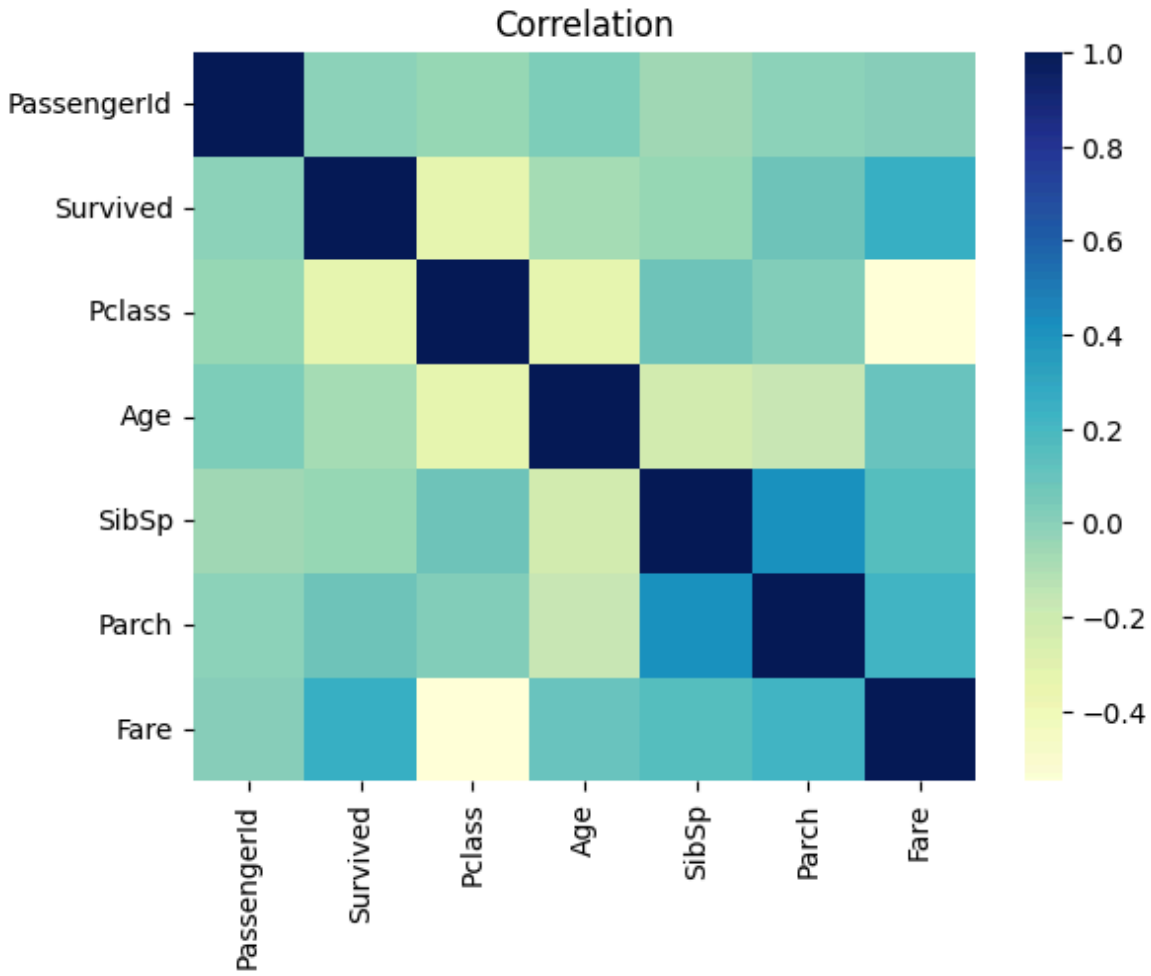
In [79]:
```
sns.jointplot(x = "Survived", y = "Fare", kind = "scatter", data = data)
```

Out[79]:
```
<seaborn.axisgrid.JointGrid at 0x1cea8616f90>
```



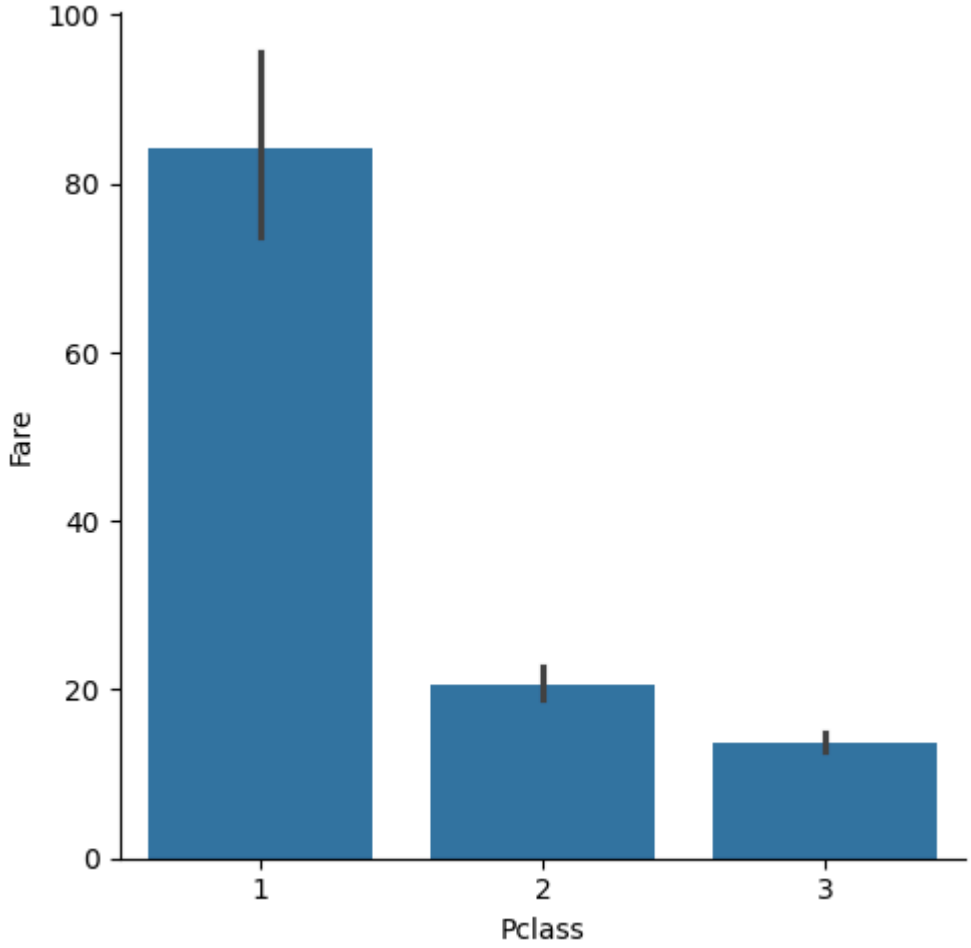
In [81]:
```
tc = data.corr(numeric_only=True)
sns.heatmap(tc, cmap="YlGnBu")
plt.title('Correlation')
```

Out[81]:
```
Text(0.5, 1.0, 'Correlation')
```



In [82]:
```
sns.catplot(x='Pclass', y='Fare', data=data, kind='bar')
```

Out[82]:
```
<seaborn.axisgrid.FacetGrid at 0x1cebd2f9e50>
```



In [ ]: